

DDA5001 Machine Learning

Training versus Testing (Part II)

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Recap: In-sample Error versus Out-of-sample Error

- **In-sample Error:** Given a set of **training samples** $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$,

$$\text{Er}_{\text{in}} = \frac{1}{n} \sum_{i=1}^n e(f(\mathbf{x}_i), g(\mathbf{x}_i))$$

- **Out-of-sample Error:** Suppose data $\mathbf{x}$ follows a certain distribution $\mathcal{D}$ in an i.i.d. manner,

$$\text{Er}_{\text{out}} = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [e(f(\mathbf{x}), g(\mathbf{x}))]$$

Remarks:

- The In-sample error Er_{in} is also known as the **training error**.
- The out-of-sample error Er_{out} is **more general than the test error**.
Fortunately, we can use the test error to approximate Er_{out} very well when the test dataset is large enough.

Recap: Concept of Training versus Testing / Generalization

Recall that learning is all about to infer g outside of the seen training dataset, i.e.,

Make the out-of-sample error small

- ▶ But Er_{out} is **not computable** at all.

Here is a simple decomposition:

$$Er_{out} = \underbrace{Er_{out} - Er_{in}}_{\text{generalization error}} + \underbrace{Er_{in}}_{\text{training error}}$$

- ▶ We need to simultaneously make generalization error and training error small, in order to make Er_{out} small.

The goal of generalization is to

explore how out-of-sample error is related to in-sample error, i.e., bound the generalization error

- ▶ The reason to explore this relationship is that Er_{in} is **computable, checkable, and even amenable** (can be directly minimized).

Recap: The Starting Point

Given training samples $\mathcal{S} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$.

In expectation for any $f \in \mathcal{H}$: (we omit θ in f_θ for simplicity)

$$\mathbb{E}_{\mathcal{S} \sim i.i.d. \mathcal{D}} [\text{Er}_{\text{in}}(f)] = \text{Er}_{\text{out}}(f).$$

Proof: Recall that all samples are i.i.d. according to $\mathcal{D}$, we have

$$\begin{aligned}\mathbb{E}_{\mathcal{S} \sim i.i.d. \mathcal{D}} [\text{Er}_{\text{in}}(f)] &= \mathbb{E}_{\mathcal{S} \sim i.i.d. \mathcal{D}} \left[\frac{1}{n} \sum_{i=1}^n e(f(\mathbf{x}_i), g(\mathbf{x}_i)) \right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [e(f(\mathbf{x}), g(\mathbf{x}))] \quad (\text{since i.i.d.}) \\ &= \text{Er}_{\text{out}}(f)\end{aligned}$$

- Law of large numbers: When $n \rightarrow \infty$, we have in-sample error estimates the out-of-sample error accurately.
- However, this is true only when $n \rightarrow \infty$.

Agenda

Finite Hypothesis Space Generalization

VC Dimension

VC Dimension-induced Generalization

Additional Assumption: Finite Hypothesis Space

Assumption: Finite hypothesis space

The cardinality $|\mathcal{H}| < +\infty$

where $|\mathcal{H}|$ measures the number of all possible $f_{\theta} \in \mathcal{H}$.

- ▶ It means that the number of possible f_{θ} in $\mathcal{H}$ is finite. Is it a practical assumption?
- ▶ We will omit θ in f_{θ} in the sequel to ease notation. Remind yourself f is almost always parameterized by some parameter θ .

Generalization for Fixed f : A Lemma

Lemma: High probability bounds for fixed f

Fix any model $f : \mathcal{X} \mapsto \{-1, 1\}$ ($f \in \mathcal{H}$ is fixed). The following inequalities hold for any $t > 0$:

$$\Pr \left[\text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f) \geq t \right] \leq e^{-2nt^2},$$

and

$$\Pr \left[\text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f) \leq -t \right] \leq e^{-2nt^2}.$$

Thus, we have the two side tail probability bound

$$\Pr \left[|\text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f)| \geq t \right] \leq 2e^{-2nt^2}.$$

- ▶ **Non-asymptotic** bounds valid for any n .
- ▶ Equivalently, $\Pr \left[|\text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f)| \leq t \right] \geq 1 - 2e^{-2nt^2}$, which is a high probability bound.

Proof

Recall the Hoeffding's inequality:

Hoeffding's inequality for bounded random variables

Suppose X_i are **independent** random variables with mean μ_i and bounded on $[a_i, b_i]$ for $i = 1, \dots, n$, then for any $t > 0$, we have

$$\Pr \left[\sum_{i=1}^n (X_i - \mu_i) \geq t \right] \leq e^{-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}}$$

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Note that $e(f(\mathbf{x}_i), g(\mathbf{x}_i))$ equals to either 0 or 1, we have

$$\begin{aligned} & \Pr \left[\text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f) \geq t \right] \\ &= \Pr \left[\frac{1}{n} \sum_{i=1}^n (e(f(\mathbf{x}_i), g(\mathbf{x}_i))) - \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n e(f(\mathbf{x}_i), g(\mathbf{x}_i)) \right] \geq t \right] \\ &= \Pr \left[\sum_{i=1}^n (e(f(\mathbf{x}_i), g(\mathbf{x}_i))) - \mathbb{E}[e(f(\mathbf{x}_i), g(\mathbf{x}_i))] \geq nt \right] \\ &\leq e^{-2nt^2}. \end{aligned}$$

Proof

Letting $\text{Er}_{\text{in}}(f) = -\text{Er}_{\text{in}}(f)$, following exactly the same argument, we get the other side

$$\Pr \left[\text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f) \leq -t \right] \leq e^{-2nt^2}$$

By **union bound**, we have

$$\begin{aligned} \Pr \left[|\text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f)| \geq t \right] &= \Pr \left[\text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f) \geq t \right. \\ &\quad \text{or } \left. \text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f) \leq -t \right] \\ &\stackrel{\text{union bound}}{\leq} 2e^{-2nt^2} \end{aligned}$$

Lemma: Union bound

For events $A_1, \dots, A_n$,

$$\Pr \left[\bigcup_{i=1}^n A_i \right] \leq \sum_{i=1}^n \Pr[A_i]$$

Generalization for Fixed Model f

Proposition: Generalization for **fixed f**

Fix a model $f : \mathcal{X} \mapsto \{-1, 1\}$. For any $\delta > 0$, the following generalization bound holds **with probability at least $1 - \delta$**

$$\text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2n}}$$

Proof: Letting $\delta = 2e^{-2nt^2}$ and solving for t yields the desired result.

Generalization for Fixed Model f

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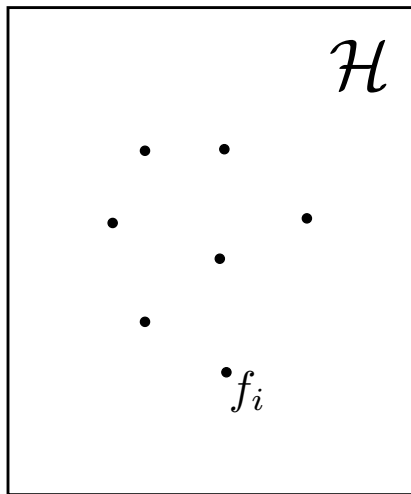
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Proof: Letting $\delta = 2e^{-2nt^2}$ and solving for t yields the desired result.

Are we done?

What We Need is Uniform Bound for All Possible $f \in \mathcal{H}$



Generalization for Finite Hypothesis Space

Theorem: Generalization for **finite hypothesis space**

Let $\mathcal{H}$ be a **finite** hypothesis space. For any $\delta > 0$, the following generalization bound holds **with probability at least $1 - \delta$**

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \sqrt{\frac{\log\left(\frac{2|\mathcal{H}|}{\delta}\right)}{2n}} \quad (1)$$

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- ▶ The dependence on δ is only **logarithmically**.
- ▶ The generalization error increases when $|\mathcal{H}|$ grows, but only **logarithmically**.
- ▶ **More samples (larger n) lead to better generalization.**

Trade-off

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \sqrt{\frac{\log\left(\frac{2|\mathcal{H}|}{\delta}\right)}{2n}}$$

On the **training** side, we need

more complex hypothesis $\mathcal{H}$ (larger $|\mathcal{H}|$)

On the **generalization** side, we need

less complex hypothesis $\mathcal{H}$ (smaller $|\mathcal{H}|$)

Proof

The proof is done by applying union bound. Let $f_1, \dots, f_{|\mathcal{H}|}$ be the elements of $\mathcal{H}$. We have

$$\begin{aligned} & \Pr \left[\exists f \in \mathcal{H} \text{ s.t. } |\text{Er}_{\text{in}}(f) - \text{Er}_{\text{out}}(f)| \geq t \right] \\ &= \Pr \left[|\text{Er}_{\text{in}}(f_1) - \text{Er}_{\text{out}}(f_1)| \geq t, \text{ or } \dots \right. \\ & \quad \left. \text{or } |\text{Er}_{\text{in}}(f_{|\mathcal{H}|}) - \text{Er}_{\text{out}}(f_{|\mathcal{H}|})| \geq t \right] \\ & \stackrel{\text{union bound}}{\leq} \sum_{i=1}^{|\mathcal{H}|} \Pr \left[|\text{Er}_{\text{in}}(f_i) - \text{Er}_{\text{out}}(f_i)| \geq t \right] \\ & \leq 2|\mathcal{H}|e^{-2nt^2} \end{aligned}$$

Setting the RHS probability to δ yields the desired result.

Issue with Finite Hypothesis Space

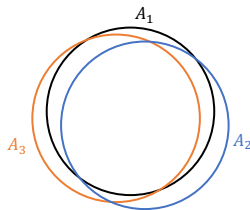
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Issue with Finite Hypothesis Space

However, $|\mathcal{H}|$ is usually infinite in practice

- ▶ Think about the the simplest (one-dimensional) linear model $f_{\theta}(x) = \theta x$, $\theta \in \mathbb{R}$. We have infinitely many choices for θ .
- ▶ If we let $|\mathcal{H}|$ tends to ∞ , we will have a trivial bound: Some finite thing $< \infty$.

Issue: Union bound can be very coarse.



In this case, $\Pr[A_1 \text{ or } A_2 \text{ or } A_3]$ is much less than $\sum_{i=1}^3 \Pr[A_i]$.

Solution: Find a way to measure the complexity of $\mathcal{H}$ more smartly.

Agenda

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VC Dimension-induced Generalization

Road Map

- ▶ Instead of using $|\mathcal{H}|$ directly to count the complexity of $\mathcal{H}$, we have to properly account for the overlaps of different hypotheses.
- ▶ In this way, our goal is to replace the number of hypotheses $|\mathcal{H}|$ with an effective number **which is finite** even when $|\mathcal{H}|$ is infinite.

Road Map

- ▶ Instead of using $|\mathcal{H}|$ directly to count the complexity of $\mathcal{H}$, we have to properly account for the overlaps of different hypotheses.
- ▶ In this way, our goal is to replace the number of hypotheses $|\mathcal{H}|$ with an effective number **which is finite** even when $|\mathcal{H}|$ is infinite.

↪ This quantity will be the so-called **growth function**, which is of **combinatorial nature**.

- ▶ The growth function captures how different the hypotheses in $\mathcal{H}$ are, and hence how much overlap the different hypotheses have.
- ▶ After that, we will introduce the notion of **VC dimension**, which is used to bound the growth function.
- ▶ Finally, we will show that we can replace $|\mathcal{H}|$ in the obtained generalization bound with VC dimension.

Before that, let us introduce the related notion called **dichotomy**.

Dichotomy

- ▶ If $f \in \mathcal{H}$ is applied to a **finite** sample $\{x_1, \dots, x_n\}$, we get n -tuple $\{f(x_1), \dots, f(x_n)\}$ of ± 1 's.
- ▶ Such a n tuple is called a **dichotomy** since it splits $\{x_1, \dots, x_n\}$ into **two** groups: The points for which f is $+1$ and those for which f is -1 .
- ▶ Each $f \in \mathcal{H}$ generates a dichotomy on $\{x_1, \dots, x_n\}$, **but two different f 's may generate the same dichotomy.**

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We can now define the dichotomies of the whole hypothesis space $\mathcal{H}$.

Dichotomies of $\mathcal{H}$

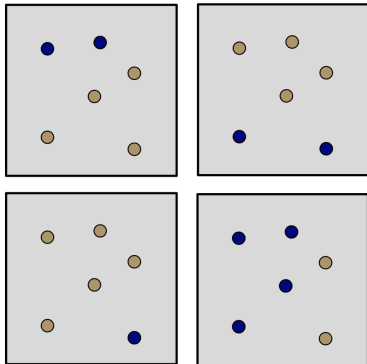
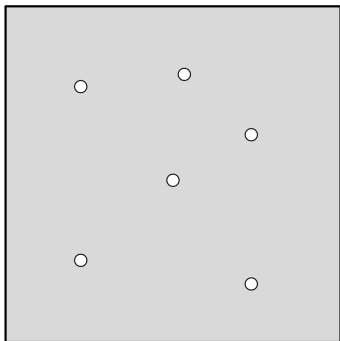
Given $\{x_1, \dots, x_n\}$, the dichotomies generated by $\mathcal{H}$ on these points are defined by

$$\mathcal{H}(x_1, \dots, x_n) = \{(f(x_1), \dots, f(x_n)) : f \in \mathcal{H}\}.$$

- ▶ One can think of the dichotomies $\mathcal{H}(x_1, \dots, x_n)$ as a set of hypothesis just like $\mathcal{H}$ is, except that the hypotheses are only seen through the n data points.
- ▶ Larger $\mathcal{H}(x_1, \dots, x_n)$ means that $\mathcal{H}$ is more diverse / rich.

Example of Dichotomies

We have six points. There are four different $f \in \mathcal{H}$ on the points.



Growth Function

Growth function is a **number**, which is defined based on the number of dichotomies.

Growth function

The growth function for the hypothesis set $\mathcal{H}$ is defined as:

$$\mathcal{G}_{\mathcal{H}}(n) = \max_{\{x_1, \dots, x_n\} \subseteq \mathcal{X}} |\mathcal{H}(x_1, \dots, x_n)|,$$

where $|\cdot|$ denotes the cardinality (number of elements) of a set.

Instead of counting the size of $\mathcal{H}$ by $|\mathcal{H}|$, the idea of growth function is:
Using $\mathcal{H}$, what is the maximum number of ways can we label a n -points dataset?

Properties of Growth Function

- ▶ $\mathcal{G}_{\mathcal{H}}(n)$ counts the **most dichotomies** that can possibly be generated on **any** n points in $\mathcal{X}$.
- ▶ To compute $\mathcal{G}_{\mathcal{H}}(n)$, we consider all possible choice of n points, and pick the one that gives us the most dichotomies, which is of **combinatorial** nature.
- ▶ Similar to $|\mathcal{H}|$, $\mathcal{G}_{\mathcal{H}}(n)$ is a measure of the **richness** of the hypothesis set $\mathcal{H}$. The difference is that it is now considered on n points rather than the entire input space $\mathcal{X}$.

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- ▶ Similar to $|\mathcal{H}|$, $\mathcal{G}_{\mathcal{H}}(n)$ is a measure of the **richness** of the hypothesis set $\mathcal{H}$. The difference is that it is now considered on n points rather than the entire input space $\mathcal{X}$.
- ▶ Since $\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_n) \subset \{-1, +1\}^n$ (the set of all possible dichotomies on any n points). Clearly, we have

$$\mathcal{G}_{\mathcal{H}}(n) \leq 2^n.$$

- ▶ If $\mathcal{H}$ is capable of generating all possible dichotomies on $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, then $\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_n) = \{-1, +1\}^n$, i.e., $\mathcal{G}_{\mathcal{H}}(n) = 2^n$, and we say that $\mathcal{H}$ can **shatter** the data points $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$.

Vapnik-Chervonenkis (VC) Dimension

We now introduce a well known notion — Vapnik-Chervonenkis (VC) dimension.

VC dimension

The VC dimension of a hypothesis space $\mathcal{H}$, denoted by $d_{\text{VC}}(\mathcal{H})$ or simply d_{VC} , is the largest n so that it can be shattered by $\mathcal{H}$, i.e.,

$$d_{\text{VC}}(\mathcal{H}) := \max\{n : \mathcal{G}_{\mathcal{H}}(n) = 2^n\}.$$

If $\mathcal{G}_{\mathcal{H}}(n) = 2^n$ for all n , then $d_{\text{VC}}(\mathcal{H}) = \infty$.

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- ▶ By definition, VC dimension indicates the representation power of $\mathcal{H}$.
- ▶ It counts the number of data points n that $\mathcal{H}$ start to not shatter.

Fact:

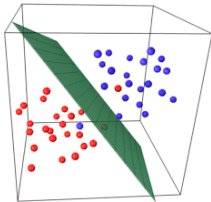
Bounding Growth Function using VC Dimension

$$\mathcal{G}_{\mathcal{H}}(n) \leq n^{d_{VC}} + 1.$$

↪ Next lecture: VC dimension generalization.

Example: VC Dimension of Linear Classifier

For a linear classifier, we can derive its VC dimension in a general sense.
This can be generalized to the following general result:



Theorem

For d -dimensional (binary) linear classifier, we have

$$d_{VC} = d + 1.$$

► Proof is put in the supplementary material.

Property of VC Dimension of Linear Classifier

- ▶ d_{VC} is exactly the number of parameters of a d -dimensional binary linear classifier (think about perceptron), i.e., $\theta_0, \theta_1, \dots, \theta_d$.
- ▶ d_{VC} measures the **effective number of parameters**, and hence the complexity of $\mathcal{H}$.
- ▶ The more parameters a model has, the more complex $\mathcal{H}$ is. This is reflected by a large d_{VC} .
- ▶ In some other models, the effective parameters may be less obvious.

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VC Dimension Generalization Result

After introducing all the related notions, we can now introduce the VC dimension generalization result.

VC generalization bound

For any $\delta > 0$, with probability at least $1 - \delta$, we have the following generalization bound:

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \sqrt{\frac{8}{n} \log \left(\frac{4\mathcal{G}_{\mathcal{H}}(2n)}{\delta} \right)}$$

Upon invoking the upper bound on growth function using VC dimension, we have

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \sqrt{\frac{8}{n} \log \left(\frac{4((2n)^{d_{\text{vc}}} + 1)}{\delta} \right)}$$

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► See the supplementary material for a proof sketch.

VC Generalization versus Previous Ones

- ▶ The VC generalization bound has the form

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \mathcal{O} \left(\sqrt{\frac{d_{\text{VC}}}{n}} \right)$$

where $\mathcal{O}$ is used to hide a $\sqrt{\log n / \delta}$ term and some constants.

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- ▶ Comparing the VC generalization bound to the finite $\mathcal{H}$ bound, it is easy to see that we not only replace $|\mathcal{H}|$ with $\mathcal{G}_{\mathcal{H}}$, but also change some constants. This is due to some technical issues. Fortunately, the overall idea is still maintained, that is, we use a much more reasonable effective number ($\mathcal{G}_{\mathcal{H}}$ or d_{VC}) to measure the complexity of $\mathcal{H}$ rather than using $|\mathcal{H}|$.

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- ▶ Larger n means that Er_{in} will generalize better to Er_{out} . When $n \rightarrow \infty$, we have $\text{Er}_{\text{in}} = \text{Er}_{\text{out}}$, which is consistent with our observation from the law of large numbers.

Is VC Generalization Bound Meaningful / Useful?

- ▶ The VC analysis is a **universal** result since it applies to all hypothesis space, learning algorithm, input space, probability distributions, binary targets (It can be extended to other target functions as well).
- ▶ Due to such a generality, **the bound is loose**.

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Though it is a little bit loose, it gives us useful guidance when conducting machine learning.

- ▶ It formally establishes the feasibility of learning for **infinite** $\mathcal{H}$. For $\mathcal{H}$ with finite d_{VC} , once we have enough training samples, learning is likely to be feasible.
- ▶ It tends to be **equally loose** for different models, enabling us to compare different models by comparing their d_{VC} . In real applications, model with smaller d_{VC} tend to generalize better than that with larger d_{VC} .
- ▶ It gives us some rules of thumb, e.g., about the number of training samples: $n \geq 10 \times d_{VC}$.

↪ We list several applications / guidance of the VC bound in practice.

Sample Complexity

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Suppose we want the result to hold with probability at least $1 - \delta$, and generalization error (error between Er_{in} and Er_{out}) to be less than some small number ε , we have

$$n \geq \frac{8}{\varepsilon^2} \log \left(\frac{4((2n)^{d_{\text{vc}}} + 1)}{\delta} \right).$$

Concisely, we need $n \geq \mathcal{O}\left(\frac{d_{\text{vc}} \log(1/\delta) \log n}{\varepsilon^2}\right)$.

Example: Estimating the Sample Complexity

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Suppose that we have a learning model with $d_{VC} = 3$ and would like the generalization error to be at most 0.1 with confidence 90% (so $\varepsilon = 0.1, \delta = 0.1$). How big a data set do we need?

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$$n \geq \frac{8}{0.1^2} \log \left(\frac{4((2n)^3 + 1)}{0.1} \right).$$

Solving the above inequality gives $n \approx 22000$. $\square$

- ▶ This obtained sample complexity is much bigger than the previously said rule of thumb $n \geq 10 \times d_{VC}$, due to the fact that VC bound is quite loose.
- ▶ Nonetheless, the practical guidance is illustrated. With larger d_{VC} , we need more samples. This is consistent with practice.